

TAITA TAVETA NATIONAL POLYTECHNIC
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING
MECHATRONIC TECHNICIAN

PAD SEGREGATOR AND INCINERATION SYSTEM

BY

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**THIS PROJECT IS SUBMITTED IN PARTIAL FULFILMENT OF THE
REQUIREMENT FOR THE AWARD OF THE DIPLOMA IN MECHATRONIC
TECHNICIAN AT TAITA TAVETA NATIONAL POLYTECHNIC**

SUPERVISOR: MASIKA STEPHANIE

YEAR: 2026

DECLARATION PAGE

I hereby declare that this project report is based on my original work except for citations and quotations which have been duly acknowledged. I also declare that it has not been previously submitted for any other academic award.

Name: **Madzikwato Elisha Jilo**

Signature: _____

Date: _____

SUPERVISOR APPROVAL

I, the undersigned, certify that this project report has been prepared under my supervision and is submitted with my approval.

Supervisor: **Masika Stephanie**

Signature: _____

Date: _____

DEDICATION PAGE

This project is dedicated to my parents, guardians, and friends for their encouragement, support, and motivation throughout my studies and during the completion of this project.

ACKNOWLEDGEMENT PAGE

I would like to express my sincere appreciation to my supervisor Ms. Masika Stephanie for her guidance, encouragement, and support throughout the development of this project.

I also extend my appreciation to the lecturers and technical staff of the Department of Electrical and Electronics Engineering at Taita Taveta National Polytechnic for the knowledge and practical skills they have provided during my course.

Finally, I thank my family and friends for their encouragement and moral support throughout the project period.

ABSTRACT

Proper management of sanitary waste is essential for maintaining hygiene and environmental safety, especially in institutions such as schools, hospitals, and public washrooms. Improper disposal of sanitary pads can lead to environmental pollution, unpleasant odors, and health risks to sanitation workers.

This project presents the design and development of a Pad Segregator and Incineration System, an automated solution designed to improve the hygienic disposal of sanitary pads. The system uses an ultrasonic sensor to detect when a user approaches the disposal unit. An LCD display guides the user by displaying instructions to insert the sanitary pad into the system.

Once the pad is inserted, a motor-driven shredding mechanism activates to shred the pad into smaller pieces. After shredding, the user presses a fire activation button, which triggers an incineration process for approximately 10 seconds to burn the shredded material safely. Once the burning process is completed, the system automatically resets and becomes ready for the next operation.

The system integrates sensor technology, microcontroller programming, mechanical shredding, and thermal disposal, demonstrating the application of mechatronic principles. The Pad Segregator and Incineration System offers a hygienic, efficient, and environmentally friendly method for managing sanitary waste in institutions.

TABLE OF CONTENTS

Contents

DECLARATION PAGE	2
SUPERVISOR APPROVAL.....	2
DEDICATION PAGE.....	3
ACKNOWLEDGEMENT PAGE.....	4
ABSTRACT	5
TABLE OF CONTENTS	6
CHAPTER ONE	8
1.1 Introduction.....	8
1.2 Background of the Study	9
1.3 Problem Statement	10
1.4 Objectives	11
General Objective	11
1.5 Scope and Limitations	12
CHAPTER TWO	13
LITERATURE REVIEW.....	13
2.1 Introduction.....	13
2.2 Sanitary Waste Management	13
2.3 Waste Incineration Technology	13
2.4 Automation in Waste Management.....	13
CHAPTER THREE.....	14
METHODOLOGY.....	14
CHAPTER FOUR.....	15
SYSTEM ANALYSIS AND REQUIREMENT MODELING	15
4.1 Introduction.....	15
4.2 Analysis of the Current System	15
4.2.1 Current Disposal Process	15
4.2.2 Problems Associated with the Current System	16
4.3 Data Collection Methods	16
4.3.1 Observation	16
4.3.2 Literature Review	17
4.3.3 Consultation	17
4.4 Requirement Analysis	17
4.5 Functional Requirements.....	17

4.6 Non-Functional Requirements	18
4.7 Proposed System Description	19
4.8 System Block Diagram.....	19
4.9 System Operation Flow	20
CHAPTER FIVE	21
SYSTEM DESIGN	21
CHAPTER SIX.....	22
SYSTEM IMPLEMENTATION	22
6.1 Introduction.....	22
6.2 Hardware Implementation	22
6.2.1 Ultrasonic Sensor Installation	23
6.2.2 LCD Display Installation	23
6.2.3 Shredding Motor Mechanism.....	23
6.2.4 Fire Activation System	24
6.2.5 Power Supply System.....	24
6.3 Software Implementation.....	24
6.4 System Operation	25
6.5 System Testing	25
6.5.1 Sensor Testing	25
6.5.2 Motor Testing.....	25
6.5.3 Incineration Testing.....	26
6.5.4 System Integration Testing	26
6.6 Changeover Strategy	26
CHAPTER SEVEN.....	27
LIMITATIONS, CONCLUSIONS AND RECOMMENDATIONS.....	27
Conclusion	27
Recommendations	27
REFERENCES	28
APPENDICES	29
Appendix A – System Block Diagram.....	29
Appendix B – Circuit Diagram.....	30
Appendix C – Flowchart	31
Appendix D – Arduino Code.....	34

CHAPTER ONE

1.1 Introduction

Menstrual hygiene management is an important public health concern, particularly in schools, hospitals, and workplaces. Sanitary pads are widely used by women for menstrual protection. However, improper disposal of used sanitary pads can lead to serious hygiene and environmental challenges.

In many institutions, sanitary pads are disposed of in ordinary waste bins or flushed in toilets. These methods often lead to blocked drainage systems, unpleasant odors, and contamination risks for waste handlers.

With advancements in automation and mechatronics, it is possible to develop systems that improve sanitary waste management through automation. Automated waste disposal systems can reduce human contact with waste while improving hygiene and environmental safety.

The Pad Segregator and Incineration System is designed to provide a safe and automated method for disposing of sanitary pads. The system uses an ultrasonic sensor to detect when a user approaches the unit and an LCD display to guide the user. After the pad is inserted, a motor-driven shredding mechanism reduces the pad into smaller pieces. The shredded material is then safely burned using a controlled incineration process.

The integration of sensors, microcontrollers, motors, and thermal components demonstrates the practical application of mechatronics engineering in solving real-world problems.

1.2 Background of the Study

Sanitary waste management has become an important issue in modern society due to increased awareness of hygiene and environmental protection. In many developing countries, the disposal of sanitary pads is often neglected due to lack of proper waste management systems.

Used sanitary pads contain biological materials that may harbor harmful microorganisms. If not disposed of properly, they can cause health hazards, environmental pollution, and unpleasant odors.

Incineration has been widely used as a method for disposing of medical and sanitary waste because it destroys harmful microorganisms and reduces the volume of waste significantly.

However, traditional incinerators often require manual handling of waste. This exposes users and sanitation workers to health risks.

The development of an automated pad segregator with incineration capability helps solve this problem by reducing human contact and ensuring hygienic waste disposal.

1.3 Problem Statement

Improper disposal of sanitary pads is a major challenge in many institutions. The common problems associated with sanitary waste disposal include:

- Mixing sanitary pads with general waste.
- Health risks to sanitation workers.
- Unpleasant odors in washrooms.
- Environmental pollution caused by improper disposal.
- Lack of automated disposal systems.

These challenges highlight the need for an automated system that can safely collect, shred, and dispose of sanitary pads hygienically.

1.4 Objectives

General Objective

To design and develop an automated Pad Segregator and Incineration System for hygienic sanitary waste disposal.

Specific Objectives

1. To design a detection system using an ultrasonic sensor.
2. To implement an LCD display to guide users during disposal.
3. To develop a motor-driven shredding mechanism for sanitary pads.
4. To implement a controlled burning system for safe waste disposal.
5. To integrate all components using a microcontroller-based control system.

1.5 Scope and Limitations

Scope

The project focuses on the design and development of a prototype automated system that:

- Detects sanitary pad disposal
- Shreds the pad
- Burns the shredded material
- Resets automatically for the next operation

Limitations

The system may be limited by:

- Power supply limitations
- Prototype size constraints
- Temperature control limitations
- Sensor accuracy

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter reviews previous research related to sanitary waste disposal, waste segregation systems, and automated waste management technologies.

2.2 Sanitary Waste Management

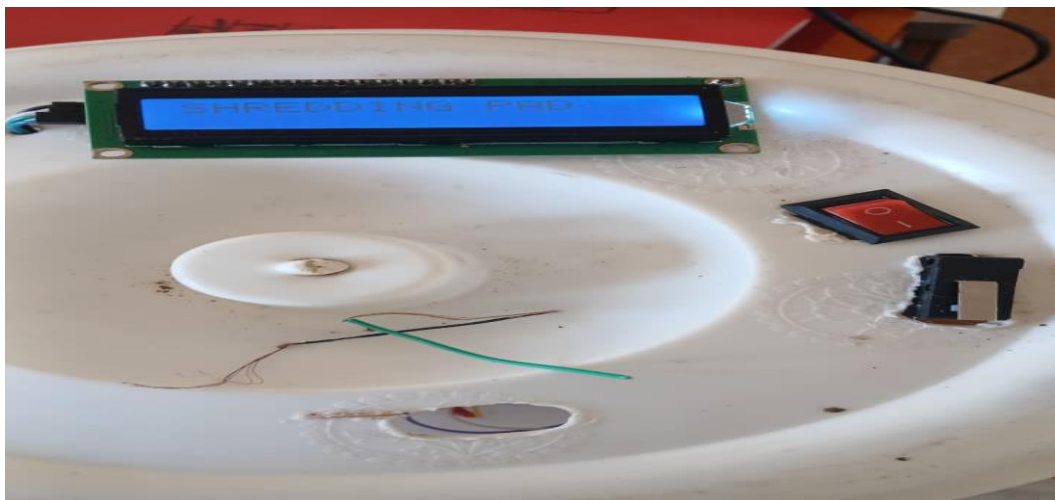
Sanitary waste refers to waste generated from personal hygiene products such as sanitary pads, diapers, and medical waste.

2.3 Waste Incineration Technology

Incineration is a process that uses high temperatures to burn waste materials. It reduces waste volume and eliminates harmful microorganisms.

2.4 Automation in Waste Management

Automation technologies such as sensors, microcontrollers, and actuators are increasingly used in waste management systems.



CHAPTER THREE

METHODOLOGY

First, the system design was developed to define how the pad segregator would operate. The design included selecting appropriate components such as the ultrasonic sensor, LCD display, motor, burner mechanism, and microcontroller. A block diagram and flowchart were created to illustrate the working process of the system.

Second, the hardware components were assembled. The ultrasonic sensor was installed to detect the presence of sanitary pads when they are inserted into the device. A motor was connected to a shredding mechanism to break down the pads into smaller pieces. A burner unit was also installed to safely incinerate the shredded pads.

Third, the software programming was developed using a microcontroller platform. The program was written to control the operations of the system. When the ultrasonic sensor detects a pad, the LCD displays instructions to the user. The motor then activates to shred the pad. After shredding, the system prompts the user to press the fire button, which activates the burner for approximately ten seconds to destroy the waste.

Finally, the system testing and evaluation were conducted to ensure that all components worked correctly. Each part of the system was tested individually and then as a complete system to verify accurate detection, shredding, burning, and automatic resetting for the next operation.

This methodology ensured that the pad segregator system functions effectively in improving hygiene and safe disposal of sanitary waste.

CHAPTER FOUR

SYSTEM ANALYSIS AND REQUIREMENT MODELING

4.1 Introduction

This chapter presents the analysis of the existing sanitary waste disposal system and identifies the requirements for the proposed Pad Segregator and Incineration System. System analysis is an important stage in the development of any mechatronic system because it helps identify the problems in the current system and determine the requirements needed to develop an improved solution.

The chapter also explains how data and information about the current system were collected and analyzed. Based on the findings, the requirements for the proposed automated system are defined. System modeling tools such as flowcharts and block diagrams can also be used to illustrate how the proposed system will function.

4.2 Analysis of the Current System

The current method of disposing sanitary pads in many institutions such as schools, hospitals, and public toilets is mainly manual. In most cases, used sanitary pads are disposed of in ordinary waste bins or flushed into toilets.

This method presents several challenges that affect hygiene, environmental safety, and waste management efficiency.

4.2.1 Current Disposal Process

The typical sanitary pad disposal process currently used in many institutions involves the following steps:

1. The user removes the sanitary pad.
2. The used pad is wrapped using paper or plastic material.
3. The wrapped pad is thrown into a normal waste bin or toilet.
4. Waste collectors later remove the waste manually.
5. The waste is transported to a landfill or disposal site.

4.2.2 Problems Associated with the Current System

The current system presents several challenges, including:

Poor Hygiene

Manual disposal of sanitary pads exposes users and waste handlers to contamination risks. Used pads contain blood and biological materials that may carry harmful microorganisms.

Environmental Pollution

Sanitary pads are made from materials that are not easily biodegradable. When disposed of improperly, they contribute to environmental pollution.

Unpleasant Odors

Waste bins containing sanitary pads often produce unpleasant smells due to bacterial decomposition.

Blocked Drainage Systems

In some cases, users flush sanitary pads in toilets. This causes blockages in plumbing systems and increases maintenance costs.

Lack of Waste Segregation

Sanitary pads are usually mixed with other waste materials, making waste management more difficult.

4.3 Data Collection Methods

To better understand the existing waste disposal system, several data collection methods were used.

4.3.1 Observation

Observation was conducted in washrooms and waste disposal areas to study how sanitary pads are currently disposed of. It was observed that most disposal systems rely on simple waste bins with no automation or segregation.

4.3.2 Literature Review

Information was also obtained from research articles, books, and online resources related to waste management, sanitary waste disposal, and automated waste systems.

4.3.3 Consultation

Consultation with lecturers and technical experts helped identify possible solutions and design approaches for the proposed system.

4.4 Requirement Analysis

After analyzing the current system, the requirements for the proposed Pad Segregator and Incineration System were identified. These requirements define what the system should be able to do.

System requirements are generally divided into two categories:

1. Functional requirements
2. Non-functional requirements

4.5 Functional Requirements

Functional requirements describe the specific tasks that the system must perform.

The proposed Pad Segregator system must perform the following functions:

Detection of User Presence

The system should detect when a user approaches the disposal unit. This is achieved using an ultrasonic sensor.

Display Instructions to the User

An LCD display should provide instructions to guide the user on how to dispose of the sanitary pad.

Accept and Store the Pad

The system should provide an opening through which the user can insert the sanitary pad.

Shredding Mechanism

Once the pad is inserted, a motor-driven shredding mechanism should shred the pad into smaller pieces.

Incineration Process

The system should include a burning or heating mechanism to incinerate the shredded pad.

Fire Activation Control

A fire activation button should be provided to initiate the incineration process for a specified duration (approximately 10 seconds).

Automatic System Reset

After the burning process is completed, the system should automatically reset and become ready for the next operation.

4.6 Non-Functional Requirements

Non-functional requirements describe the performance and operational characteristics of the system.

These include:

Safety

The system must be designed to ensure safe operation, especially during the burning process.

Reliability

The system should operate consistently without frequent failures.

Ease of Use

Users should be able to use the system easily without requiring technical knowledge.

Energy Efficiency

The system should minimize energy consumption during operation.

Durability

The mechanical and electronic components should be durable and capable of long-term operation.

4.7 Proposed System Description

The proposed Pad Segregator and Incineration System is an automated waste management device designed to improve the hygienic disposal of sanitary pads.

The system integrates the following technologies:

- Ultrasonic sensor for detecting user presence
- LCD display for user guidance
- Motorized shredding mechanism
- Fire activation system for incineration
- Microcontroller for system control

4.8 System Block Diagram

The system can be represented using a block diagram showing the relationship between different components.

The main components include:

1. Ultrasonic Sensor
2. Microcontroller Unit
3. LCD Display
4. Motor Driver
5. Shredding Motor
6. Fire Activation Button
7. Heating Element / Burner
8. Power Supply

These components work together to ensure the system operates efficiently.

4.9 System Operation Flow

The operation of the system follows the following sequence:

1. The ultrasonic sensor detects a user approaching the system.
2. The LCD displays instructions to insert the sanitary pad.
3. The user inserts the pad into the system.
4. The microcontroller activates the shredding motor.
5. The pad is shredded into smaller pieces.
6. The user presses the fire activation button.
7. The system activates the heating element for approximately 10 seconds.
8. The shredded pad is burned.
9. The system resets automatically and becomes ready for the next operation.

CHAPTER FIVE

SYSTEM DESIGN

The system consists of the following components:

- Ultrasonic sensor
- LCD display
- Microcontroller
- Motor-driven shredding mechanism
- Fire activation button
- Heating element for burning
- Control circuit

CHAPTER SIX

SYSTEM IMPLEMENTATION

6.1 Introduction

This chapter describes the implementation of the Pad Segregator and Incineration System.

System implementation refers to the process of constructing, programming, assembling, and testing the designed system to ensure that it performs according to the intended specifications.

The implementation stage involved assembling the electronic components, developing the control program using a microcontroller, constructing the mechanical structure, and performing various tests to verify that the system operates correctly. The main goal of this stage was to transform the system design into a functional prototype capable of detecting sanitary pads, shredding them, and safely burning the shredded material.

6.2 Hardware Implementation

The hardware implementation involved assembling all the electronic and mechanical components required for the system. These components were interconnected according to the circuit design developed in the previous chapter.

The main hardware components used in the system included:

Ultrasonic sensor

LCD display

Microcontroller (Arduino Uno)

Motor driver module

DC motor for shredding

Heating element or burner

Fire activation push button

Power supply unit

Protective housing structure

6.2.1 Ultrasonic Sensor Installation

The ultrasonic sensor was installed near the opening of the pad segregator unit. The sensor was used to detect the presence of a user approaching the system. When the sensor detected an object within a predefined distance, a signal was sent to the microcontroller to activate the display instructions on the LCD.

The sensor was connected to the microcontroller using trigger and echo pins, which allowed the system to measure distance and detect the presence of an object.

6.2.2 LCD Display Installation

An LCD display was installed on the front panel of the system to guide the user during the disposal process. The display showed messages such as:

“Pad segregator system welcome”

“Place sanitary pad in the hole”

“Shredding pad”

“Hold combustion Button 10 seconds”

The LCD was connected to the microcontroller using digital communication pins, allowing the microcontroller to send text messages to the display.

6.2.3 Shredding Motor Mechanism

A DC motor was used to drive the shredding mechanism inside the system. The motor was connected to a set of cutting blades designed to shred the sanitary pad into smaller pieces.

The motor was controlled through a motor driver circuit that allowed the microcontroller to activate or stop the motor when required.

When a pad was detected inside the system, the microcontroller activated the motor for a short duration to shred the pad before the incineration stage.

6.2.4 Fire Activation System

A push button switch was installed on the control panel to allow the user to initiate the burning process. When the button was pressed, the microcontroller activated the heating element or burner for approximately ten seconds.

This controlled burning process ensured that the shredded sanitary pad was completely destroyed.

6.2.5 Power Supply System

The system required a stable power supply to operate the electronic components and the motor. A regulated power supply unit was used to provide the required voltage to the microcontroller, sensors, and other electronic components.

Separate power regulation was used for high-power components such as the motor and heating element to prevent damage to the microcontroller.

6.3 Software Implementation

The software implementation involved writing a control program that enabled the microcontroller to coordinate the operation of all system components.

The program was developed using the Arduino Integrated Development Environment (IDE).

The software controlled the following operations:

Reading signals from the ultrasonic sensor

Displaying instructions on the LCD

Activating the shredding motor

Controlling the burning process

Resetting the system after operation

The program was uploaded to the Arduino microcontroller through a USB connection from a computer.

6.4 System Operation

The implemented system operated according to the following sequence:

The ultrasonic sensor detected the presence of a user approaching the disposal unit.

The LCD display instructed the user to insert the sanitary pad.

Once the pad was inserted, the microcontroller activated the shredding motor.

The motor shredded the pad into smaller pieces.

The LCD display instructed the user to press the fire button.

When the fire button was pressed, the heating element activated for approximately ten seconds.

The shredded material was burned during this period.

After the burning process was completed, the system automatically reset and returned to standby mode.

6.5 System Testing

After implementation, the system was tested to verify that all components functioned correctly. Testing ensured that the system met the design requirements and operated safely.

6.5.1 Sensor Testing

The ultrasonic sensor was tested to confirm that it could detect objects within the required distance range. The sensor successfully detected a user approaching the system and triggered the LCD display.

6.5.2 Motor Testing

The shredding motor was tested to ensure that it could rotate and operate the cutting blades effectively. The motor successfully shredded the inserted material.

6.5.3 Incineration Testing

The burning system was tested to confirm that the heating element could activate for the required duration and burn the shredded material effectively.

6.5.4 System Integration Testing

All components were tested together to verify that the entire system operated smoothly and followed the correct operational sequence.

6.6 Changeover Strategy

The proposed system could replace traditional sanitary waste bins in washrooms and public sanitation facilities. The implementation of the system would involve installing the unit in appropriate locations and training users on how to operate it safely.

The automated system would reduce manual waste handling and improve hygiene in waste disposal areas.

CHAPTER SEVEN

LIMITATIONS, CONCLUSIONS AND RECOMMENDATIONS

Conclusion

The Pad Segregator and Incineration System provides a hygienic and efficient solution for sanitary pad disposal. By integrating automation and controlled incineration, the system improves hygiene and reduces environmental pollution.

Recommendations

Future improvements may include:

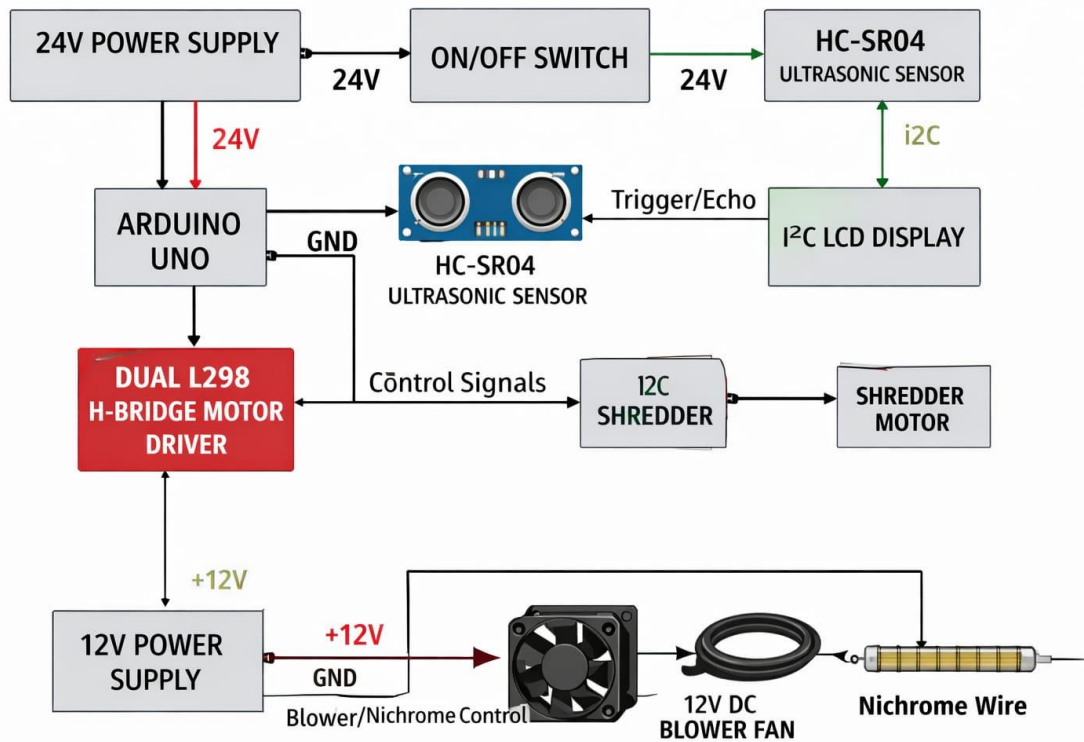
- Automatic incineration without manual button activation
- IoT monitoring of waste levels
- Solar-powered operation
- Improved safety sensors

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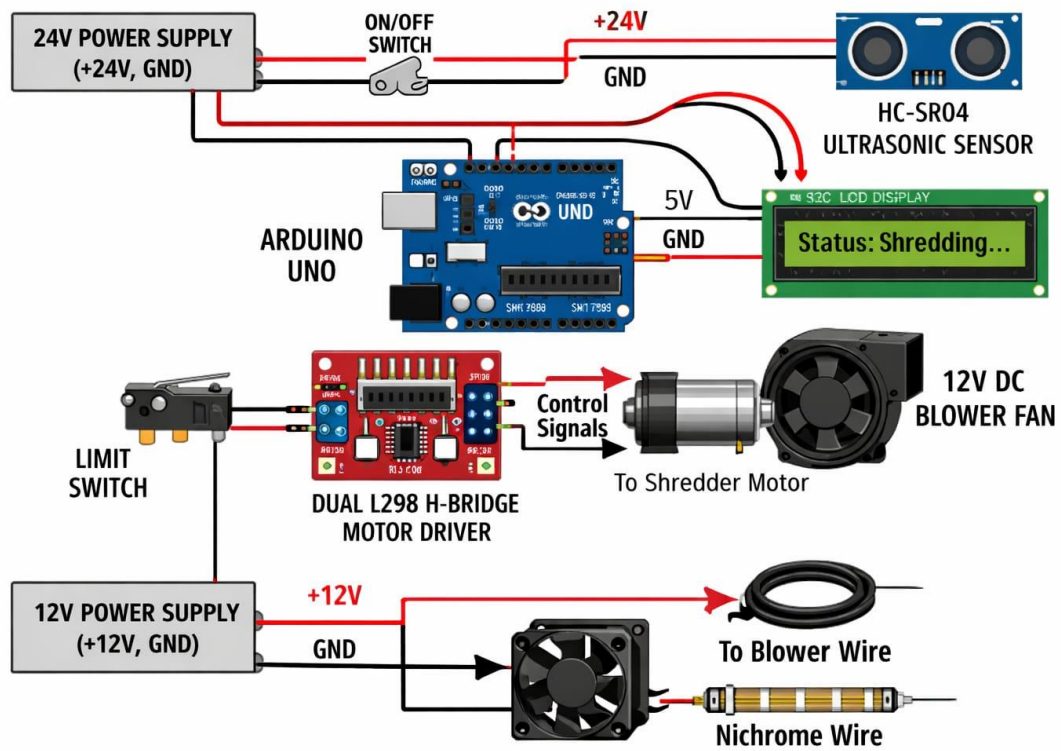
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APPENDICES

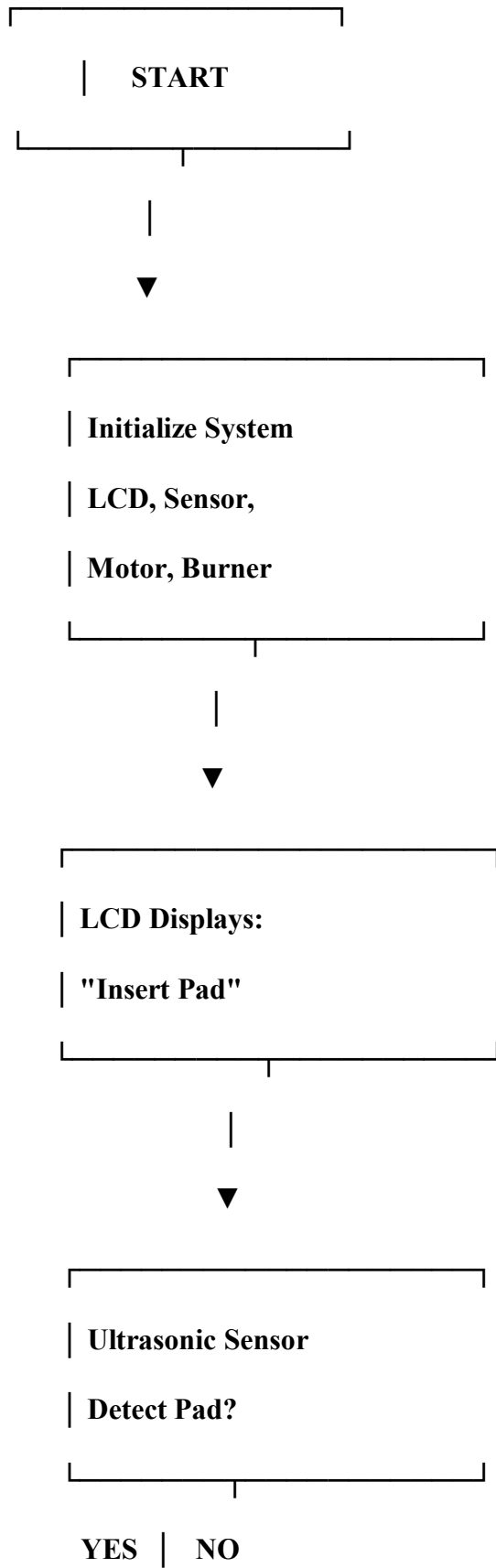
Appendix A – System Block Diagram

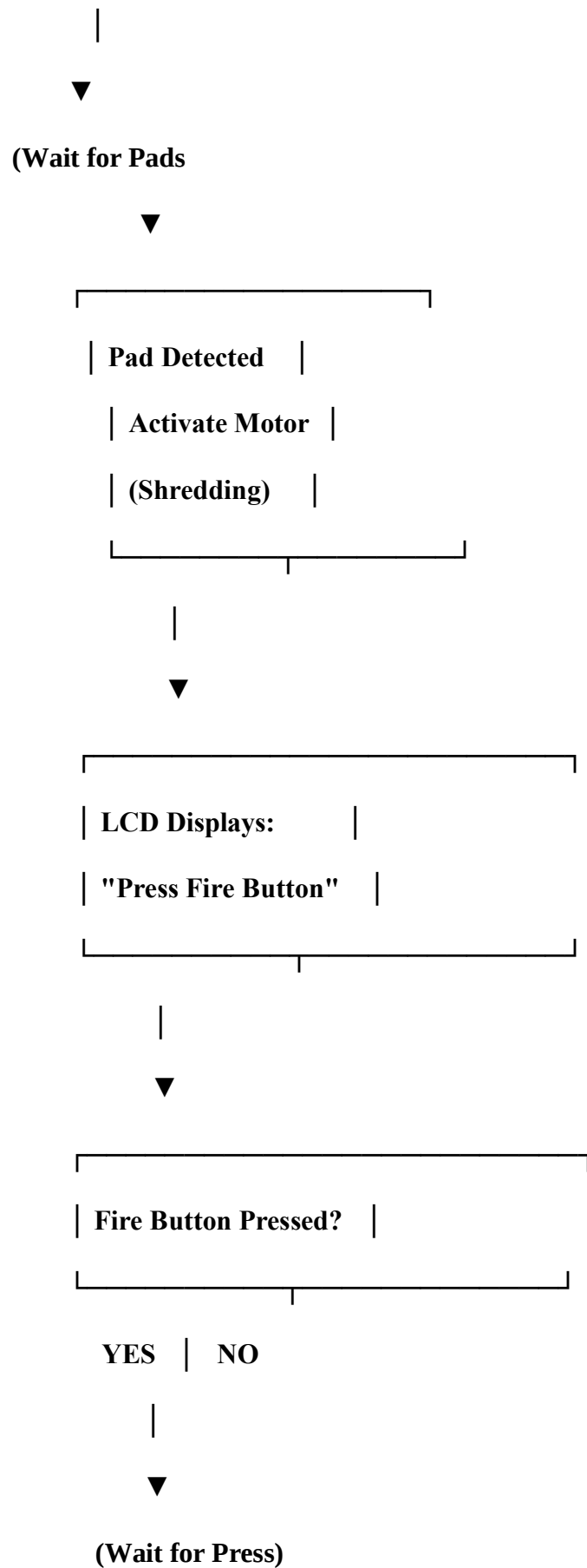


Appendix B – Circuit Diagram



Appendix C – Flowchart





Appendix D – Arduino Code

```
#include <Wire.h>
```

```
#include <LiquidCrystal_I2C.h>
```

```
LiquidCrystal_I2C lcd(0x3F, 16, 2);
```

```
// Ultrasonic pins
```

```
const int trigPin = 9;
```

```
const int echoPin = 10;
```

```
// Motor driver pins
```

```
const int IN1 = 7;
```

```
const int IN2 = 6;
```

```
const int ENA = 5;
```

```
// Distance thresholds (adjust during calibration)
```

```
int triggerDistance = 15; // pad detected
```

```
int clearDistance = 18; // chamber empty
```

```
void setup()
```

```
{
```

```
  pinMode(trigPin, OUTPUT);
```

```
  pinMode(echoPin, INPUT);
```

```
  pinMode(IN1, OUTPUT);
```

```

pinMode(IN2, OUTPUT);

pinMode(ENA, OUTPUT);


lcd.init();

lcd.backlight();


lcd.setCursor(0,0);

lcd.print("PAD SEGREGATOR");

lcd.setCursor(0,1);

lcd.print("SYSTEM WELCOME");

delay(3000);


lcd.clear();

}


void loop()

{

int distance = readDistance();


// IDLE STATE

if(distance > triggerDistance)

{

lcd.setCursor(0,0);

lcd.print("PLACE SANITARY ");

```

```

    lcd.setCursor(0,1);

    lcd.print("PAD IN THE HOLE");

    delay(500);

}

```

```

// PAD DETECTED

if(distance <= triggerDistance)

{

    lcd.clear();

    lcd.setCursor(0,0);

    lcd.print("SHREDDING PAD");

```

```

// Start motor

```

```

digitalWrite(IN1,HIGH);

digitalWrite(IN2,LOW);

analogWrite(ENA,200);

```

```

// Keep shredding until chamber empty

```

```

while(true)

{

    distance = readDistance();

    if(distance > clearDistance)

    {

```

```

        delay(800); // confirm empty

        if(readDistance() > clearDistance)

            break;
    }

    delay(200);
}

// Stop motor

digitalWrite(IN1,LOW);

digitalWrite(IN2,LOW);

analogWrite(ENA,0);

lcd.clear();

lcd.setCursor(0,0);

lcd.print("HOLD COMBUSTION");

lcd.setCursor(0,1);

lcd.print("BUTTON 10 SECS");

delay(5000);

lcd.clear();
}
}

```

```

// Stable ultrasonic reading function

int readDistance()

{

    int total = 0;

    for(int i=0;i<3;i++)

    {

        digitalWrite(trigPin, LOW);

        delayMicroseconds(2);


        digitalWrite(trigPin, HIGH);

        delayMicroseconds(10);

        digitalWrite(trigPin, LOW);


        long duration = pulseIn(echoPin, HIGH, 30000);


        int distance = duration * 0.034 / 2;


        total += distance;


        delay(50);

    }


    return total / 3;

}

```